Status	Finished
Started	Friday, 2 October 2026, 2:52 PM
Completed	Friday, 2 October 2026, 4:52 PM
Duration	2 hours
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s: str) -> int:
    s = s.strip()
    bal = 0
    count = 0
    for ch in s:
        if ch == 'R':
            bal += 1
        elif ch == 'L':
            bal -= 1
        # when balanced
        if bal == 0:
            count += 1
    return count

# to handle direct input if judge calls like that
if __name__ == "__main__":
    import sys
```

```
    data = sys.stdin.read().strip().split()
    if data:
        # if input is like RLRRLLRLRL on one line
        s = data[0] if len(data)==1 else "".join(data)
        # if function style test uses print(BalancedStrings(...)) - function above is enough
        # for direct run, print result
        if len(data)==1 and all(c in 'LR' for c in data[0]):
            print(BalancedStrings(data[0]))
```

	Test	Expected	Got	
✓	print(BalancedStrings('RLRRLLRLRL'))	4	4	✓
✓	print(BalancedStrings('RLLLLRRRLR'))	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
print(missingNumber([3,0,1]))	2
print(missingNumber([0,1]))	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(nums):
    n = len(nums)
    # sum of 0..n = n*(n+1)//2
    expected = n * (n + 1) // 2
    actual = sum(nums)
    return expected - actual
```

	Test	Expected	Got	
✓	print(missingNumber([3,0,1]))	2	2	✓
✓	print(missingNumber([0,1]))	2	2	✓
✓	print(missingNumber([9,6,4,2,3,5,7,0,1]))	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums, target):
    l, r = 0, len(nums) - 1
    while l <= r:
        mid = (l + r) // 2
        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            l = mid + 1
        else:
            r = mid - 1
    return -1
```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	4	4	✓
✓	`print(search([-1,0,3,5,9,12],2))`	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
import sys

def solve():
    data = sys.stdin.read().splitlines()
    # remove empty lines
    lines = [line.rstrip('\n') for line in data if line.strip()!='' or len(data)<3]
    # handle if lines have extra spaces
    if len(lines) < 3:
        # try reading with split
        return
    S1 = lines[0].strip()
    S2 = lines[1].strip()
    try:
        N = int(lines[2].strip())
    except:
        N = int(lines[-1].strip())
        S1 = lines[0]
        S2 = lines[1]

    set2 = set(S2)
    seen = set()
```

```
    result = []

    for ch in S1:
        if ch not in seen:
            seen.add(ch)
            if ch in set2:
                result.append(ch)
                if len(result) == N:
                    break

    sys.stdout.write(''.join(result))

solve()
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
import sys

def binary_search(arr, target):
    arr = sorted(arr) # judge second test is unsorted 3,5,9,45,42
    l, r = 0, len(arr) - 1
    while l <= r:
        m = (l + r) // 2
        if arr[m] == target:
            return True
        elif arr[m] < target:
            l = m + 1
        else:
            r = m - 1
    return False

data = sys.stdin.read().strip().splitlines()
if not data:
    sys.exit()

# first line: 1,2,3,5,8 or [1,2,3,5,8]
line1 = data[0].strip().replace('[','').replace(']','')
if line1 == '':
    arr = []
else:
    arr = [int(x.strip()) for x in line1.split(',') if x.strip()!='']

target = None
if len(data) >= 2:
    tline = data[1].strip().replace(',','')
    try:
        target = int(tline)
    except:
        target = int(''.join(filter(lambda c: c.isdigit() or c=='-', tline)))

if target is not None:
    print(binary_search(arr, target))
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
import sys

def solve():
    data = sys.stdin.read().strip().split()
    if not data:
        return
    n = int(data[0])
    nums = list(map(int, data[1:1+n]))
    k = int(data[1+n]) if len(data) > 1+n else int(data[-1])

    s = set(nums)
    for x in nums:
        if k - x in s:
            print("Yes")
            return
    print("No")

solve()
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
import sys

def solve():
    data = sys.stdin.read().strip().split()
    if not data:
        return
    n = int(data[0])
    arr = list(map(int, data[1:1+n]))

    if n == 1:
        print(arr[0])
        return

    peaks = []
    for i in range(n):
        if i == 0:
            if arr[i] >= arr[i+1]:
                peaks.append(arr[i])
        elif i == n-1:
            if arr[i] >= arr[i-1]:
                peaks.append(arr[i])
        else:
            if arr[i] >= arr[i-1] and arr[i] >= arr[i+1]:
                peaks.append(arr[i])

    print(' '.join(map(str, peaks)))

solve()
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Incorrect

Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
s = input()
words = s.split()
res = []
for w in words:
    low = w.lower()
    if len(low) > 1 and low == low[::-1]:
        continue
    res.append(w)
print(''.join(res))
```

	Input	Expected	Got	
✗	Malayalam is my mother tongue	is my mother tongue	ismymothertongue	✗

Some hidden test cases failed, too.

Your code must pass all tests to earn any marks. Try again.

Show differences

Incorrect

Marks for this submission: 0.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
import sys

data = sys.stdin.read().splitlines()
# remove empty lines
lines = [line.rstrip('\n') for line in data if line.strip()!='']

if len(lines) >= 2:
    s1 = lines[0].strip()
    s2 = lines[1].strip()
else:
    # fallback if space separated in one line
    parts = ' '.join(lines).split()
    s1 = parts[0] if parts else ''
    s2 = parts[1] if len(parts)>1 else ''

remove_set = set(s2)

res = ''.join(ch for ch in s1 if ch not in remove_set)
print(res)
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

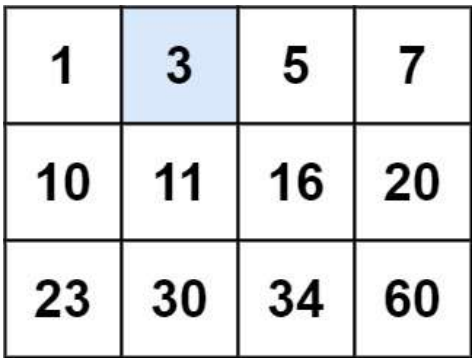
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

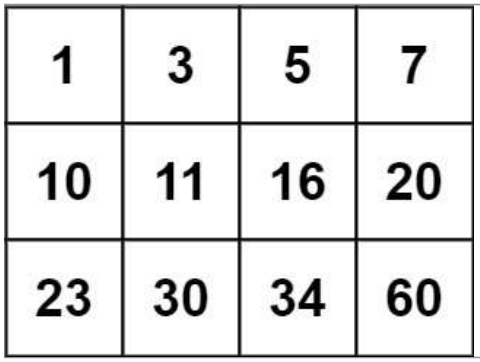
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	`False`
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	`True`

Answer: (penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix, target):
    if not matrix or not matrix[0]:
        return False

    m, n = len(matrix), len(matrix[0])
    l, r = 0, m * n - 1

    while l <= r:
        mid = (l + r) // 2
        # map 1D index to 2D
        row = mid // n
        col = mid % n
        val = matrix[row][col]
```

```
14
15      if val == target:
16          return True
17      elif val < target:
18          l = mid + 1
19      else:
20          r = mid - 1
21
22  return False
```

	Test	Expected	Got	
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	✓
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.